is not always the same. Traffic pattern, thus, keeps changing continuously. There is a drastic change in the number of users at day time than at night time. Due to such variations in traffic pattern, there arises a demand for a flexible technology that should be capable of handling such traffic.

To eradicate vendor dependence. Traditional networking devices are functioned and programmed by technicians. The network administrator cannot modify the program or update the protocol. Thus, when any new algorithm or protocol is to be implemented in the network, altogether, the existing devices become useless.

Evolution of Cloud technology. Companies are now moving towards Cloud computing. Most of their data is stored on Cloud servers for easy and quick access. Ultimately, it increases the burden on network architecture and increases traffic.

Increasing number of networking devices connected through the Internet. As shown in Fig. 2, researchers predict from the statistics that the number of connected networking devices may reach 50 billion by 2020.

Fig. 3 reveals the complexity of the network. To manage and control such a tremendous volume of traffic, appropriate architecture is required.

Evolution of the Internet of Things (IoT). As shown in Fig. 4, the IoT may connect 50.1 billion networking devices by 2020. The IoT market revenue is expected to grow from US$ 1.928 billion in 2013 to US$ 7.065 billion in 2020. Such a rapid evolution results in the demand for highly-reliable and agile networking architecture. Data-intensive services offered by the IoT like video surveillance, emergency services and uninterrupted monitoring demand a technology that supports the cause with excellent quality of service and high data rates.

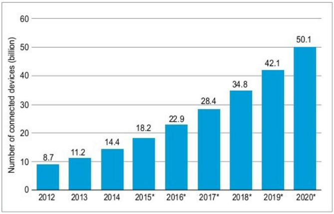

Fig. 2: Number of connected networking devices worldwide (Courtesy: www.statista.com)

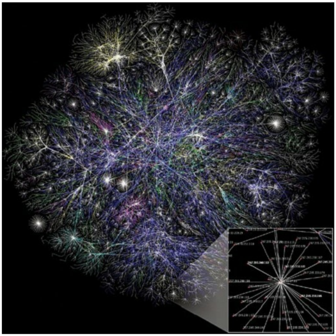

Fig. 3: Complexity of network due to connected networking devices (Image courtesy: https://en.wikipedia.org)

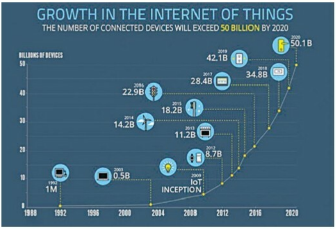

Fig. 4: Growth of the IoT

Architecture components of SDN

Fig. 5 shows the various components of SDN architecture. The architecture comprises three planes, namely, data, control and application. Data plane includes the dedicated hardware and is responsible for the processing of packets as well as traffic forwarding.

In the controller plane, policies are configured to define the scope of control given to SDN application and to monitor system performance. Optimising resource allocation and checking network failure are also the roles of controllers. In both planes, security functions are considered to ensure secured intercommunication.

As demonstrated in Fig. 6, in traditional networking, each device has an embedded set of protocols that make the routing decision. Whereas in SDN (Fig. 7), there is a central network operating system (OS) that controls all devices in the network. All decision-making protocols